Exp No: 4 **Date:**	**Integer Programming Problem- Branch and bound method**

AIM:

To solve any integer programming problem using the Branch and Bound method in Python.

PROCEDURE:

1. Formulate the integer programming problem.
2. Implement the Branch and Bound method in Python.
3. Solve the problem using the implemented method.

Problem-1:

Maximize

$Z = 6x + 9y$

Subject to

$\mathbf{3x + 2y \leqslant 18}$

$\mathbf{x + 4y \leqslant 20}$

$\mathbf{x + y \leqslant 9}$

$\mathbf{x \geqslant 0 , y \geqslant 0}$

Code:

```
from scipy.optimize import

linprog c = [-6, -9]


A = [
        [3, 2],
        [1, 4],
        [1, 1]
]



b = [18, 20, 9]



x_bounds = (0,
None) y_bounds
= (0, None)

result = linprog(c, A_ub=A, b_ub=b, bounds=[x_bounds, y_bounds])

print("LP Relaxation Solution:", result.x)



    print("Objective Value:", -result.fun)
```

Output:

```
LP Relaxation Solution: [2. 4.]
Objective Value: 48.0
```

RESULT:

Thus, the Dynamic Programming technique was successfully executed and verified using Python.